Lab 1 Outline

1. Introduction

A. Societal Problem

1. The beverage manufacturing industry relies on spreadsheets, email chains, phone calls, and disconnected software systems to coordinate production activities.
2. As beverage companies grow, these manual processes create communication delays, approval bottlenecks, and operational inefficiencies.
3. The beverage packaging market was valued at approximately $157.7 billion in 2023 and is projected to reach $222.1 billion by 2030 (Grand View Research, 2024).
4. Beverage manufacturers need a centralized system that improves collaboration across suppliers, production teams, logistics providers, and clients.
5. The complexity of beverage manufacturing and packaging investments increase the need for coordinated, scalable information systems that help organizations manage production activity across multiple stakeholders (Zimmerman, 2026).

B. Problem and Needed Solution Characteristics

1. Centralized workflow management: organizes operational activities within one platform and reduces dependence on scattered spreadsheets and email.
2. Real-time visibility into production activities: allow authorized stakeholders to view current requests, approvals, schedules, production progress, and delivery status.
3. Automated approval tracking: record responsibility and status while reducing delays caused by repeated manual follow-up.
4. Production scheduling: coordinates milestones and helps teams recognize delays before they affect delivery.
5. Logistics coordination: connects production completion, shipping activity, and delivery information.
6. Document management: stores operational files in an organized location where authorized users can locate the current version.
7. Reporting and dashboard capabilities: turn operational data into useful information for oversight and decision making.
8. Secure role-based user access: limits users to the information and actions appropriate to their responsibilities.
9. Scalability to support business growth: support additional users, customers, workflows, and business growth without replacing the operational foundation.

C. Product Introduction

1. BevOS is a centralized workflow management platform designed specifically for beverage manufacturers.
2. The system connects brands, suppliers, distributors, co-packers, logistics providers, and internal operations through a single platform.
3. BevOS improves efficiency by reducing manual coordination, improving communication, and increasing operational visibility.
4. A functional prototype will demonstrate how the product's interface, services, modules, database work together to address the identified coordination problem.

2. BevOS Product Description

BevOS is a centralized workflow management platform designed specifically for beverage manufacturing operations. The platform streamlines communication, production scheduling, approval tracking, logistics coordination, and document management by bringing key stakeholders into a single system. BevOS aims to improve operational efficiency, reduce manual processes, increase visibility throughout the production lifecycle, and support scalable business growth.

A. Purpose

1. Centralize beverage manufacturing workflow and related information.
2. Reduce dependence on spreadsheets, email chains, phone calls, and disconnected applications.
3. Create consistent coordination among brands, distributors, suppliers, production personnel, logistics providers, and finance teams.
4. Provide a shared operational view without attempting to replace every existing enterprise system or human decision-making process.

B. Goals and Objectives

1. Improve operational efficiency by reducing repeated manual coordination.
2. Reduce communication and approval delays.
3. Increase visibility throughout the production lifestyle.
4. Organize documents and operational records within their related workflows.
5. Support timely, data-informed management decisions through reports and dashboards.
6. Provide a modular foundation that can adapt as operational processes and business need evolve.

Figure 1. BevOS Objective Breakdown

C. Intended Users

1. Brands initiate product requests and monitor progress.
2. Distributors coordinate product availability and movement.
3. Suppliers receive requests, provide quote information, and support material or service needs.
4. Authorized operational personnel, including production, logistics, and finance teams, contribute information needed to complete the workflow

2.1. Key Product Features and Capabilities

A. Workflow Management

1. Centralizes operational activities and tracks work throughout the production lifestyle.
2. Connects requests, approvals, documents, communications, production, and logistics information.
3. Solves the fragmentation problem by keeping related work within one coordinated process.

B. Product Request Management

1. Allows authorized users to create, review, and monitor product requests.
2. Records request details and status within a consistent workflow.
3. Reduces the risk of incomplete requests or lost information at the beginning of a project.

C. Quote Management

1. Creates and tracks supplier quote requests and supplier resources.
2. Keeps quote information associated with the relevant product requests
3. Improves visibility and reduce repeated follow-up during supplier selection and cost evaluation.

D. Approval Tracking

1. Manages artwork, purchase-order, and other workflow approvals.
2. Records approval status, responsibility, and history while supporting notifications or escalations.
3. Reduces bottlenecks by making required actions and outstanding decisions visible.

E. Production Management

1. Coordinates production schedules, milestones, and progress updates.
2. Connects approved work to manufacturing activities.
3. Helps teams identify delays and communication changes before they affect delivery.

F. Logistics Coordination

1. Tracks shipping activity and product-delivery status.
2. Connects production completion with transportation and delivery information.
3. Provides stakeholders with a clear view of where a product is within the fulfillment process.

G. Document Management

1. Stores and organizes artwork, purchase orders, production files, shipping records, and other operational documents.
2. Associates documents with the appropriate request or workflow and restricts access to authorized users.
3. Reduces version confusion and time spent searching across separate folders or email attachments.

H. Communication Platform

1. Centralizes communication and updates among authorized stakeholders.
2. Associates communicate with the relevant request, quote, approval, production activity, or shipment.
3. Prevents important decisions from becoming separated from the work they affect.

I. Reporting and Dashboards

1. Summarizes operational status, workflow activity, delays, and other performance information.
2. Provides role-appropriate views for monitoring and decision-making.
3. Improves visibility by turning stored operation data into accessible information.

J. Issue Tracking

1. Records workflow problems, responsible partiers, status, and resolution information.
2. Uses recurring issues and user feedback to identify process weaknesses.
3. Supports contentious improvement without removing necessary human judgement.

K. Issue Tracking

1. Limits information and system actions according to each user's role and responsibilities.
2. Maintains traceable records of important workflow changes and decisions.
3. Protects confidential information while improving accountability.

2.2. Major Components (Hardware/Software)

A. Hardware Components

1. Users access BevOS from internet-connected desktop or laptop computers capable of running a modern web browser.
2. The prototype does not require specialized manufacturing hardware because BevOS coordinates information workflows rather than operating production equipment.
3. Server, database, and document-storage resources support the application software and its operational data.

B. Software Components

Figure 2. BevOS GUI Mockup

1. BevOS Web User Interface: provides the user-facing interface through which brands, distributors, and suppliers access authorized system functions.
2. BevOS Core Services: receive requests from UI, applies workflow and access rules, coordinates functional modules, and manages data exchange.
3. Quotes: manage quote requests, supplier responses, and related quote information.
4. Communications: organizes messages, notifications, and workflow-related updates.
5. Production: manages production schedules, milestones, status, and progress information.
6. Logistics: manages shipment coordination and delivery-status information.
7. Document storage: stores and retrieves operational files associated with BevOS workflows.
8. Reporting: presents operational summaries, dashboards, and decision-support information.
9. PostgreSQL: stores structured application data, including users, requests, quotes, approvals, production records, logistics records, and report data.

C. MFCD Structure and Data flow.

1. Brands, distributors, and suppliers interact with the system through BevOS UI.
2. The BevOS UI sends user requests to BevOS Core Services.
3. BevOS Core Services coordinates the production, logistics, quotes, communications, reporting, and document storage.
4. The functional modules use PostgreSQL to store and retrieve the structured information needed to complete their responsibilities.
5. This modular structure separates the user interaction, shared application logic, specialized operational functions, and persistent data storage.
6. The separation supports maintainability and scalability because modules can evolve while continuing to use common core services and data.

3. Glossary

- **Approval Workflow -** A process used to review and approve operational tasks before they move to the next stage.
- **BevOS -** A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services -** The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram) -** A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL -** An open-source relational database management system used to store application data.
- **Role-Based Access Control -** A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth -** A trusted, centralized set of current information used consistently by authorized stakeholders.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow -** A sequence of business activities completed to achieve a specific objective.

4. References

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